

Big Data Analytics Lab

Course Metadata

Credit Structure	L-T-P: 0-0-3 Total: 2
Offering Semester	Spring
Level of Subject	6 th Sem, UG Level
Compulsory or elective?	Core
Pre-requisite	Nil
Faculty Names	Jiaul H Paik, Plaban Kumar Bhowmick, Koustav Rudra
Overlapping Subjects	Nil
Justification of overlap	None

Course Description

Large data, if leveraged well, can reveal useful information. Traditional computing systems are not scalable enough to process large data. Thus, it requires parallel computing frameworks with appropriate hardware and a suitable programming framework to utilize the hardware efficiently and conveniently. This lab course introduces the programming for modern large data processing systems on scale-up as well as on scale-out systems.

Course Content

- Introduction functional and multithreaded programming
- Map-reduce Programming and its scalability issues
- Programming with Spark using RDD and structured objects
- Processing streaming data
- Processing graph data in spark with GraphX
- Large scale distributed optimization

Lecture-wise Breakup

Sl No.	Module	Unit	Module Hour (36)
1	Multithreading Programming	Multithreaded Programming & Dealing with Shared Memory	3
2	Functional Programming Basics	Functional Programming with Python	3
		Functional Programming with Scala	3
3	Map-reduce Programming	Working with Distributed File System, Map and Reduce Functions	3
		Programming with Combiner and Partitioner	3
4	Programming with Resilient	Programing with Resilient Distributed Datasets (RDD) in Spark, RDD Partitioning,	3
		Iterative Programming in Spark, RDD Persistence	3

	Distributed Datasets	Working with Distributed Structured Data in Spark: Dataframes and Datasets	3
5	Scalable Map-reduce Programming	Value-to-Key Design Pattern, Minimizing Data Communication	3
6	Stream Processing	Stream Processing with Spark Streaming, Distributed Discretized Stream, Windowed Operations, Watermarking	3
7	Graph Processing	Distributed Graph Processing with Graphx: Basic Operations on Graph, Graph Partitioning, Triangle Counting, Community Discovery	3
8	Large Scale Optimization	Distributed Stochastic Gradient Descent, Training Deep Nets with Spark Mlib	3

Books

1. **Data-Intensive Text Processing with Map-Reduce:** Lin and Dyer, Morgan and Claypool, 2010
2. **Hadoop: The Definitive Guides:** Tom White, O'REILLY 3e, 2012
3. **Spark - The Definitive Guide:** Chambers and Zaharia, O'REILLY, 2018